



Práctica 3. Logs - Parte 2

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Seguridad en el Desarrollo de Aplicaciones

Objetivo

Generar un volumen representativo de registros de eventos utilizando la aplicación VulnerableApp, analizar la información registrada mediante Seq y los archivos de log, y elaborar un reporte técnico que documente los resultados obtenidos, incluyendo la respuesta a las preguntas de análisis planteadas en la guía de práctica.

Navegación general

- 30 accesos a la página principal (Home/Index).
- 1 acceso a la vista Privacy.

Búsquedas

- 161 solicitudes al buscador (Search/Index), incluyendo:
- 120 búsquedas con términos válidos (usuario1...usuario120).
- 21 búsquedas con parámetro vacío/nulo.
- 20 búsquedas con caracteres especiales (\$%&*).
- 20 búsquedas con cadena de SQL Injection: 1' OR '1'='1.

Autenticación

- 170 intentos de inicio de sesión vía POST /Auth/Login: 150 con el usuario 'admin' y 20 con usuarios inexistentes (usuario_fantasma_1 a 20).
- Los 170 intentos quedaron registrados como 'Inicio de sesión fallido'; no se identificó en el log ningún evento de autenticación exitosa durante esta ejecución (ver observación en la pregunta 7).
- Se confirmó que la contraseña nunca se registra en el log; solo se registran el usuario y la IP.

Comentarios

- 150 intentos de creación de comentario, de los cuales 130 se registraron como 'Comentario añadido exitosamente'.
- 30 comentarios con carga XSS (<script>alert(1)</script>).
- 20 comentarios vacíos, generando advertencia (Warning).

API

- 230 solicitudes a GET /api/user/{id}: 200 con id=1 (válido) y 30 con identificadores inválidos.
- 30 solicitudes a recursos inexistentes (/api/recurso_inexistente_1 ... 30), todas con respuesta 404.
- Cada acceso no autorizado o inexistente generó un evento Warning ('Intento de acceso no autorizado...').

Excepciones

- 20 excepciones controladas registradas como Error ('Error detectado. RequestId: ...'), generadas al navegar a Home/Error tras el envío de comentarios.
- No se identificaron en este export excepciones no controladas (HTTP 500) ni mensajes 'Unhandled' provenientes del Exception Middleware; ver observación en Evidencias.

Warnings

- 422 eventos Warning en total, generados por: inicios de sesión fallidos, accesos no autorizados/IDOR, comentarios vacíos y otros mensajes internos del framework (por ejemplo, advertencias de redirección HTTPS y de consultas LINQ).

Resultados Obtenidos

Distribución de eventos por nivel

Nivel	Cantidad	% del total
Information	8,146	94.9%
Warning	422	4.9%
Error	20	0.2%
Total	8,588	100%

Eventos por controlador (según 'Executing endpoint')

Controlador	Eventos registrados
ApiController	920
AuthController	684
SearchController	644
CommentController	604
HomeController	204

Endpoints con mayor número de solicitudes

Endpoint	Solicitudes
GET /api/user/{id}	230
POST /Auth/Login	170
GET /Search/Index	161
POST /Comment/AddComment	150
GET /	30
GET /Home/Error	20

Preguntas de Análisis

1. ¿Cuántos eventos Information fueron registrados?

Se registraron 8,146 eventos de nivel Information, que representan el 94.9% del total de eventos capturados (8,588).

2. ¿Cuántos eventos Warning fueron registrados?

Se registraron 422 eventos de nivel Warning

3. ¿Cuántos eventos Error fueron registrados?

Se registraron 20 eventos de nivel Error

4. ¿Qué controlador generó más registros?

ApiController fue el controlador que generó más registros en el log, con 920 eventos asociados a la ejecución de sus endpoints, seguido de AuthController (684) y SearchController (644).

5. ¿Cuál fue el endpoint con mayor número de solicitudes?

El endpoint con mayor número de solicitudes fue GET /api/user/{id}, con 230 solicitudes registradas

6. ¿Cuál fue la dirección IP con mayor actividad?

La dirección con mayor actividad fue ::1, con 761 eventos que la incluyen explícitamente en el mensaje.

7. ¿Cuántos intentos de autenticación fallidos existieron?

Se registraron 170 intentos de autenticación fallidos en total: 150 con el usuario 'admin' y 20 con usuarios inexistentes (usuario_fantasma_1 a usuario_fantasma_20).

8. ¿Cuántos intentos de SQL Injection fueron identificados?

Se identificaron 20 intentos de SQL Injection, todos correspondientes a búsquedas en Search/Index utilizando el término 1' OR '1'='1, cumpliendo con el volumen definido en las pruebas obligatorias.

9. ¿Cuántos posibles intentos de XSS fueron registrados?

Se registraron 30 intentos de XSS, correspondientes a comentarios enviados con la carga <script>alert(1)</script> a través de Comment/AddComment.

10. ¿Cuál fue la solicitud con mayor tiempo de ejecución?

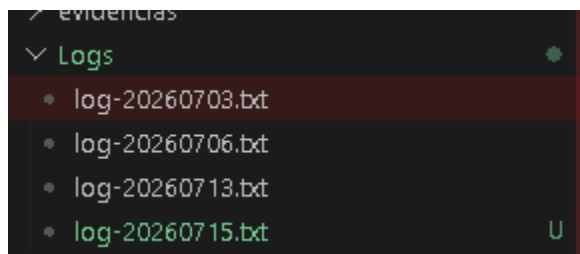
La solicitud más lenta fue GET /Search/Index?search=usuario1, con un tiempo de respuesta de 3,015.2 ms

11. ¿Fue posible localizar una petición específica utilizando únicamente el CorrelationId?

Sí, a nivel de código. CorrelationIdMiddleware genera el identificador, lo agrega como encabezado de respuesta (X-Correlation-ID) y además lo inyecta en el contexto de logging con LogContext.PushProperty("CorrelationId", cid), por lo que Serilog lo adjunta como propiedad estructurada a cada evento generado dentro de esa solicitud y lo envía al sink de Seq como propiedad indexable y filtrable (visible también en los templates de Console y de archivo).

Evidencias

1. Captura de la carpeta Logs



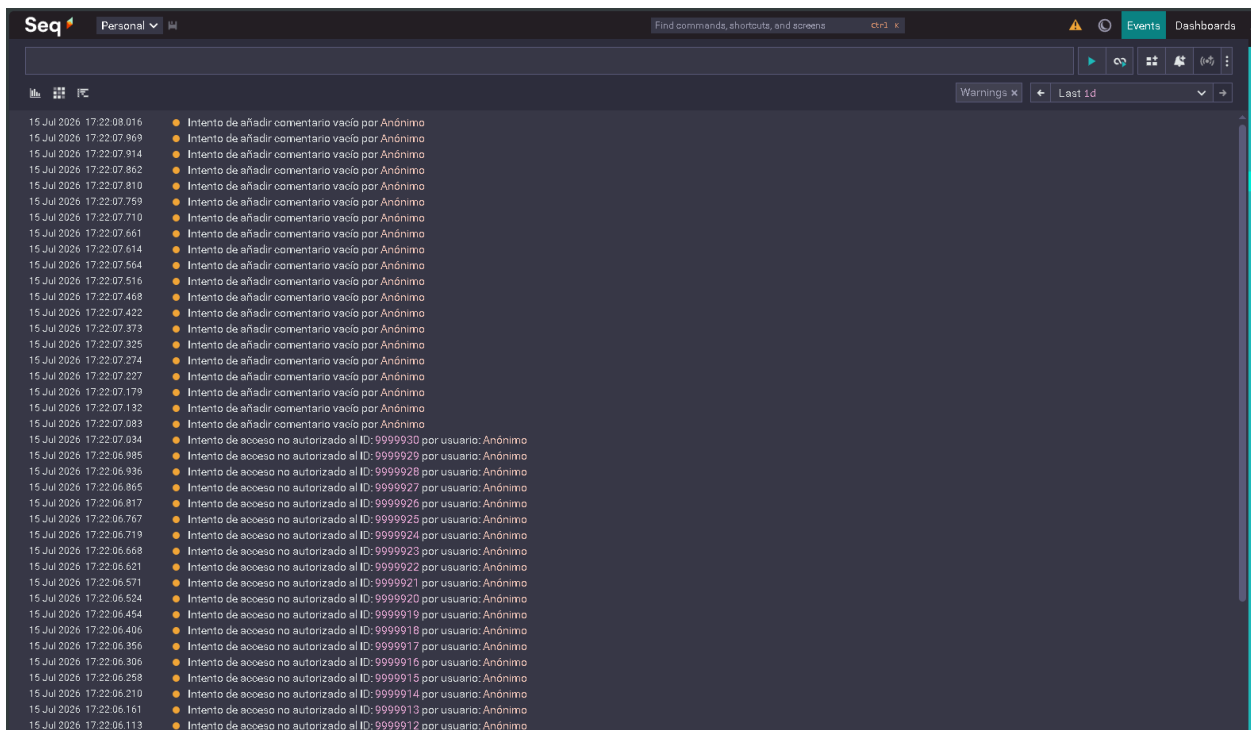
2. Captura del archivo de log generado

```
log-20260715.txt U
logs > log-20260715.txt
1 2026-07-15 16:45:12.509 -06:00 [INF] User profile is available. Using 'C:\Users\Huesos\AppData\Local\ASP.NET\DataProtection-Keys' as key repository and Windows DPAPI
to encrypt keys at rest.
2 2026-07-15 16:45:12.584 -06:00 [INF] Now listening on: http://localhost:5018
3 2026-07-15 16:45:12.587 -06:00 [INF] Application started. Press Ctrl+C to shut down.
4 2026-07-15 16:45:12.588 -06:00 [INF] Hosting environment: Development
5 2026-07-15 16:45:12.588 -06:00 [INF] Content root path: C:\Users\Huesos\Documents\VulnerableApp
6 2026-07-15 16:45:52.984 -06:00 [INF] Request starting HTTP/1.1 GET http://localhost:5018/ - null null
7 2026-07-15 16:45:52.993 -06:00 [WRN] Failed to determine the https port for redirect.
8 2026-07-15 16:45:53.017 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Executing endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
9 2026-07-15 16:45:53.040 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Route matched with {action = "Index", controller = "Home"}. Executing controller action with
signature Microsoft.AspNetCore.Mvc.IActionResult Index() on controller VulnerableApp.Controllers.HomeController (VulnerableApp).
10 2026-07-15 16:45:53.048 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Entrada a Index. Usuario: Anónimo, IP: ::1
11 2026-07-15 16:45:53.049 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Salida de Index. Tiempo: 4ms
12 2026-07-15 16:45:53.060 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Executing ViewResult, running view Index.
13 2026-07-15 16:45:53.128 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Executed ViewResult - view Index executed in 69.8992ms.
14 2026-07-15 16:45:53.130 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Executed action VulnerableApp.Controllers.HomeController.Index (VulnerableApp) in 87.6641ms
15 2026-07-15 16:45:53.132 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef Executed endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
16 2026-07-15 16:45:53.136 -06:00 [INF] 7033fdr3-6329-42f2-9a9e-e2b5a8ce75ef HTTP GET / responded 200 in 123.2512 ms
17 2026-07-15 16:45:53.140 -06:00 [INF] Request finished HTTP/1.1 GET http://localhost:5018/ - 200 null text/html; charset=utf-8 157.6892ms
18 2026-07-15 16:45:53.168 -06:00 [INF] Request starting HTTP/1.1 GET http://localhost:5018/ - null null
19 2026-07-15 16:45:53.170 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Executing endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
20 2026-07-15 16:45:53.172 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Route matched with {action = "Index", controller = "Home"}. Executing controller action with
signature Microsoft.AspNetCore.Mvc.IActionResult Index() on controller VulnerableApp.Controllers.HomeController (VulnerableApp).
21 2026-07-15 16:45:53.174 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Entrada a Index. Usuario: Anónimo, IP: ::1
22 2026-07-15 16:45:53.175 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Salida de Index. Tiempo: 0ms
23 2026-07-15 16:45:53.176 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Executing ViewResult, running view Index.
24 2026-07-15 16:45:53.178 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Executed ViewResult - view Index executed in 2.8331ms.
25 2026-07-15 16:45:53.179 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Executed action VulnerableApp.Controllers.HomeController.Index (VulnerableApp) in 5.9839ms
26 2026-07-15 16:45:53.183 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 Executed endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
27 2026-07-15 16:45:53.184 -06:00 [INF] 8f93634b-3955-417d-arb2-5695257326b7 HTTP GET / responded 200 in 14.3809 ms
28 2026-07-15 16:45:53.185 -06:00 [INF] Request finished HTTP/1.1 GET http://localhost:5018/ - 200 null text/html; charset=utf-8 18.615ms
29 2026-07-15 16:45:53.215 -06:00 [INF] Request starting HTTP/1.1 GET http://localhost:5018/ - null null
30 2026-07-15 16:45:53.218 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Executing endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
31 2026-07-15 16:45:53.219 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Route matched with {action = "Index", controller = "Home"}. Executing controller action with
signature Microsoft.AspNetCore.Mvc.IActionResult Index() on controller VulnerableApp.Controllers.HomeController (VulnerableApp).
32 2026-07-15 16:45:53.221 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Entrada a Index. Usuario: Anónimo, IP: ::1
33 2026-07-15 16:45:53.222 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Salida de Index. Tiempo: 0ms
34 2026-07-15 16:45:53.223 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Executing ViewResult, running view Index.
35 2026-07-15 16:45:53.225 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Executed ViewResult - view Index executed in 1.4346ms.
36 2026-07-15 16:45:53.226 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Executed action VulnerableApp.Controllers.HomeController.Index (VulnerableApp) in 4.4195ms
37 2026-07-15 16:45:53.227 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 Executed endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
38 2026-07-15 16:45:53.230 -06:00 [INF] b17ebcaf-40ac-4cc0-9ea9-e27c75633922 HTTP GET / responded 200 in 12.4098 ms
39 2026-07-15 16:45:53.232 -06:00 [INF] Request finished HTTP/1.1 GET http://localhost:5018/ - 200 null text/html; charset=utf-8 16.7068ms
40 2026-07-15 16:45:53.259 -06:00 [INF] Request starting HTTP/1.1 GET http://localhost:5018/ - null null
41 2026-07-15 16:45:53.261 -06:00 [INF] 1f3ac3a6-1024-4c0f-969a-428c83a1b80b Executing endpoint 'VulnerableApp.Controllers.HomeController.Index (VulnerableApp)'
42 2026-07-15 16:45:53.262 -06:00 [INF] 1f3ac3a6-1024-4c0f-969a-428c83a1b80b Route matched with {action = "Index", controller = "Home"}. Executing controller action with
signature Microsoft.AspNetCore.Mvc.IActionResult Index() on controller VulnerableApp.Controllers.HomeController (VulnerableApp).
43 2026-07-15 16:45:53.264 -06:00 [INF] 1f3ac3a6-1024-4c0f-969a-428c83a1b80b Entrada a Index. Usuario: Anónimo, IP: ::1
44 2026-07-15 16:45:53.265 -06:00 [INF] 1f3ac3a6-1024-4c0f-969a-428c83a1b80b Salida de Index. Tiempo: 1ms
45 2026-07-15 16:45:53.266 -06:00 [INF] 1f3ac3a6-1024-4c0f-969a-428c83a1b80b Executing ViewResult, running view Index.
```

3. Seq — Filtro por nivel Information

```
15 Jul 2026 17:22:09.031 Request finished HTTP/1.1 GET http://localhost:5018/Home/Error - 200 null text/html; charset=utf-8 10.0382ms
15 Jul 2026 17:22:09.030 HTTP GET /Home/Error responded 200 in 6.9646 ms
15 Jul 2026 17:22:09.029 Executed endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:09.028 Executed action 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)' in 2.6756ms
15 Jul 2026 17:22:09.027 Executed ViewResult - view Error executed in 0.9543ms.
15 Jul 2026 17:22:09.026 Executing ViewResult, running view Error.
15 Jul 2026 17:22:09.026 Error detectado. RequestId: 00-d613fbb85227746729a1bbf78e1c850ca21eff70ab29d2b4-00
15 Jul 2026 17:22:09.024 Route matched with {action = "Error", controller = "Home"}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Error() on controller VulnerableApp.Controllers.HomeController (...
15 Jul 2026 17:22:09.023 Executing endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:09.021 Request starting HTTP/1.1 GET http://localhost:5018/Home/Error - null null
15 Jul 2026 17:22:08.994 Request finished HTTP/1.1 GET http://localhost:5018/Home/Error - 200 null text/html; charset=utf-8 11.6098ms
15 Jul 2026 17:22:08.981 HTTP GET /Home/Error responded 200 in 6.6316 ms
15 Jul 2026 17:22:08.980 Executed endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:08.979 Executed action 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)' in 2.6424ms
15 Jul 2026 17:22:08.978 Executed ViewResult - view Error executed in 0.9385ms.
15 Jul 2026 17:22:08.977 Executing ViewResult, running view Error.
15 Jul 2026 17:22:08.976 Error detectado. RequestId: 00-3156cd3ae01a8f0b7016875b0cb55794-b6164fd8a292a033-00
15 Jul 2026 17:22:08.975 Route matched with {action = "Error", controller = "Home"}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Error() on controller VulnerableApp.Controllers.HomeController (...
15 Jul 2026 17:22:08.974 Executing endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:08.972 Request starting HTTP/1.1 GET http://localhost:5018/Home/Error - null null
15 Jul 2026 17:22:08.937 Request finished HTTP/1.1 GET http://localhost:5018/Home/Error - 200 null text/html; charset=utf-8 9.4615ms
15 Jul 2026 17:22:08.936 HTTP GET /Home/Error responded 200 in 6.4046 ms
15 Jul 2026 17:22:08.935 Executed endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:08.934 Executed action 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)' in 2.7741ms
15 Jul 2026 17:22:08.933 Executed ViewResult - view Error executed in 0.9653ms.
15 Jul 2026 17:22:08.932 Executing ViewResult, running view Error.
15 Jul 2026 17:22:08.931 Error detectado. RequestId: 00-446ab1694fcc0f48a9a6cb0aa79efafdee58d9e34ac03oc769-00
15 Jul 2026 17:22:08.930 Route matched with {action = "Error", controller = "Home"}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Error() on controller VulnerableApp.Controllers.HomeController (...
15 Jul 2026 17:22:08.929 Executing endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:08.927 Request starting HTTP/1.1 GET http://localhost:5018/Home/Error - null null
15 Jul 2026 17:22:08.889 Request finished HTTP/1.1 GET http://localhost:5018/Home/Error - 200 null text/html; charset=utf-8 10.9312ms
15 Jul 2026 17:22:08.888 HTTP GET /Home/Error responded 200 in 7.5237 ms
15 Jul 2026 17:22:08.887 Executed endpoint 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)'
15 Jul 2026 17:22:08.886 Executed action 'VulnerableApp.Controllers.HomeController.Error (VulnerableApp)' in 2.9027ms
```

4. Seq — Filtro por nivel Warning



5. Seq — Filtro por nivel Error



6. Seq — Búsqueda por CorrelationId

● Intento de añadir comentario vacío por Anónimo			
Event	Level (Warning)	Type (0x85041FA8)	Trace (2072...)
✓ x	ActionId	a6aa063f-81b0-467b-89bf-6c4c22bdfadf	
✓ x	ActionName	VulnerableApp.Controllers.CommentController.AddComment (VulnerableApp)	
✓ x	ConnectionId	0HNN2OV2RK43S	
✓ x	CorrelationId	c19468df-584b-4be2-847d-2fcf9d465880	
✓ x	RequestId	0HNN2OV2RK43S:00000001	
✓ x	RequestPath	/Comment/AddComment	
✓ x	SourceContext	VulnerableApp.Controllers.CommentController	
✓ x	User	Anónimo	

7. Endpoint con mayor actividad

15 Jul 2026 17:22:07.039	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999930 - 401 null application/json; charset=utf-8 10.0654ms
15 Jul 2026 17:22:07.038	HTTP GET /api/user/9999930 responded 401 in 6.9218 ms
15 Jul 2026 17:22:07.029	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999930 - null null
15 Jul 2026 17:22:06.990	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999929 - 401 null application/json; charset=utf-8 10.4851ms
15 Jul 2026 17:22:06.988	HTTP GET /api/user/9999929 responded 401 in 7.2059 ms
15 Jul 2026 17:22:06.979	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999929 - null null
15 Jul 2026 17:22:06.941	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999928 - 401 null application/json; charset=utf-8 10.4465ms
15 Jul 2026 17:22:06.940	HTTP GET /api/user/9999928 responded 401 in 7.1245 ms
15 Jul 2026 17:22:06.931	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999928 - null null
15 Jul 2026 17:22:06.870	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999927 - 401 null application/json; charset=utf-8 9.7316ms
15 Jul 2026 17:22:06.869	HTTP GET /api/user/9999927 responded 401 in 6.7180 ms
15 Jul 2026 17:22:06.860	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999927 - null null
15 Jul 2026 17:22:06.823	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999926 - 401 null application/json; charset=utf-8 11.2184ms
15 Jul 2026 17:22:06.820	HTTP GET /api/user/9999926 responded 401 in 6.6224 ms
15 Jul 2026 17:22:06.812	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999926 - null null
15 Jul 2026 17:22:06.772	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999925 - 401 null application/json; charset=utf-8 9.5444ms
15 Jul 2026 17:22:06.770	HTTP GET /api/user/9999925 responded 401 in 6.6653 ms
15 Jul 2026 17:22:06.762	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999925 - null null
15 Jul 2026 17:22:06.723	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999924 - 401 null application/json; charset=utf-8 9.4116ms
15 Jul 2026 17:22:06.722	HTTP GET /api/user/9999924 responded 401 in 6.4562 ms
15 Jul 2026 17:22:06.714	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999924 - null null
15 Jul 2026 17:22:06.673	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999923 - 401 null application/json; charset=utf-8 10.4396ms
15 Jul 2026 17:22:06.672	HTTP GET /api/user/9999923 responded 401 in 7.2577 ms
15 Jul 2026 17:22:06.663	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999923 - null null
15 Jul 2026 17:22:06.626	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999922 - 401 null application/json; charset=utf-8 9.8622ms
15 Jul 2026 17:22:06.625	HTTP GET /api/user/9999922 responded 401 in 6.4949 ms
15 Jul 2026 17:22:06.616	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999922 - null null
15 Jul 2026 17:22:06.576	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999921 - 401 null application/json; charset=utf-8 10.3491ms
15 Jul 2026 17:22:06.575	HTTP GET /api/user/9999921 responded 401 in 6.9882 ms
15 Jul 2026 17:22:06.566	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999921 - null null
15 Jul 2026 17:22:06.528	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999920 - 401 null application/json; charset=utf-8 9.9508ms
15 Jul 2026 17:22:06.527	HTTP GET /api/user/9999920 responded 401 in 6.8919 ms
15 Jul 2026 17:22:06.518	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999920 - null null
15 Jul 2026 17:22:06.459	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999919 - 401 null application/json; charset=utf-8 10.3748ms
15 Jul 2026 17:22:06.457	HTTP GET /api/user/9999919 responded 401 in 7.0460 ms
15 Jul 2026 17:22:06.448	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999919 - null null
15 Jul 2026 17:22:06.410	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999918 - 401 null application/json; charset=utf-8 10.0849ms
15 Jul 2026 17:22:06.409	HTTP GET /api/user/9999918 responded 401 in 6.9108 ms
15 Jul 2026 17:22:06.400	Request starting HTTP/1.1 GET http://localhost:5018/api/user/9999918 - null null
15 Jul 2026 17:22:06.361	Request finished HTTP/1.1 GET http://localhost:5018/api/user/9999917 - 401 null application/json; charset=utf-8 10.6445ms

8. Evento de autenticación

```
15 Jul 2026 17:21:46.136 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:46.133 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 12.9964ms
15 Jul 2026 17:21:46.118 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
15 Jul 2026 17:21:46.116 Executing endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:46.073 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:46.071 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 14.8712ms
15 Jul 2026 17:21:46.054 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
15 Jul 2026 17:21:46.053 Executing endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:46.006 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:46.004 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 10.6156ms
15 Jul 2026 17:21:45.991 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
15 Jul 2026 17:21:45.990 Executing endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.945 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.943 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 12.8345ms
15 Jul 2026 17:21:45.928 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
15 Jul 2026 17:21:45.926 Executing endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.884 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.882 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 13.2528ms
15 Jul 2026 17:21:45.866 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
15 Jul 2026 17:21:45.865 Executing endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.820 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.818 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 12.077ms
15 Jul 2026 17:21:45.804 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
15 Jul 2026 17:21:45.803 Executing endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.758 Executed endpoint 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)'
15 Jul 2026 17:21:45.757 Executed action 'VulnerableApp.Controllers.AuthController.Login (VulnerableApp)' in 11.0692ms
15 Jul 2026 17:21:45.743 Route matched with {action = 'Login', controller = 'Auth'}. Executing controller action with signature Microsoft.AspNetCore.Mvc.IActionResult Login(System.String, System.String) on controller 'VulnerableApp....
```

9. Evento Warning

```
15 Jul 2026 17:22:02.741 ● Intento de acceso no autorizado al ID: 1 por usuario: Anónimo

Event ▼ Level (Warning) ▼ Type (0x278112D2) ▼ Trace (67fd...) ▼ Export ▼
✓ x ActionId c958475e-ec3b-4f78-ac88-6d82bb3971a0
✓ x ActionName VulnerableApp.Controllers.ApiController.GetUser (VulnerableApp)
✓ x ConnectionId 0HNN2OV2RK40K
✓ x CorrelationId 4474201f-8780-4983-9f26-7a98d3263bef
✓ x Id 1
✓ x RequestId 0HNN2OV2RK40K:00000001
✓ x RequestPath /api/user/1
✓ x SourceContext VulnerableApp.Controllers.ApiController
✓ x User Anónimo
```

10. Excepción registrada

```
15 Jul 2026 17:22:08.832 ● Error detectado. RequestId: 00-0c5a87f4d2dd0d037111e8ea2b0094c5-aaff85fcb2b3ab8d-00

Event ▼ Level (Error) ▼ Type (0x3ECCE0E6) ▼ Trace (0c5a...) ▼ Export ▼
✓ x ActionId cb13cdb7-aacb-4e3f-aedc-bbe03ebc1a8f
✓ x ActionName VulnerableApp.Controllers.HomeController.Error (VulnerableApp)
✓ x ConnectionId 0HNN2OV2RK44I
✓ x CorrelationId 54940155-2500-4bd3-b794-847f99673c27
✓ x RequestId 00-0c5a87f4d2dd0d037111e8ea2b0094c5-aaff85fcb2b3ab8d-00
✓ x RequestPath /Home/Error
✓ x SourceContext VulnerableApp.Controllers.HomeController
```

11. Código fuente de Program.cs

```
C# Program.cs
You, yesterday | 2 authors (LarzenVazquez and one other)
1  using VulnerableApp.Data;
2  using Microsoft.EntityFrameworkCore;
3  using Serilog;
4  using VulnerableApp.Middleware;
5
6  var builder = WebApplication.CreateBuilder(args);
7
8  // 1.- Configuración de servicios
9  builder.Services.AddControllersWithViews();
10 builder.Services.AddSession(options =>
11 {
12     options.IdleTimeout = TimeSpan.FromMinutes(30);
13     options.Cookie.HttpOnly = true;
14     options.Cookie.IsEssential = true;
15 });
16
17 builder.Services.AddDbContext<AppDbContext>(options =>
18     options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));
19
20 Log.Logger = new LoggerConfiguration()
21     .MinimumLevel.Information()
22     .Enrich.FromLogContext()
23     .WriteTo.Console(outputTemplate: "[{Timestamp:HH:mm:ss} {Level:u3}] {CorrelationId} {Message:l}{NewLine}{Exception}")
24     .WriteTo.File("Logs/Log-.txt",
25         rollingInterval: RollingInterval.Day,
26         outputTemplate: "{Timestamp:yyyy-MM-dd HH:mm:ss.fff zzz} [{Level:u3}] {CorrelationId} {Message:l}{NewLine}{Exception}")
27     .WriteTo.Seq("http://localhost:5341")
28     .CreateLogger();
29
30 builder.Host.UseSerilog();
31
32 var app = builder.Build();
33
34 // 3.- Middlewares (EL ORDEN ES CLAVE)
35 if (!app.Environment.IsDevelopment())
36 {
37     app.UseExceptionHandler("/Home/Error");
38     app.UseHsts();
39 }
40
41 app.UseHttpsRedirection();
42 app.UseStaticFiles();
43 app.UseRouting();
44     LarzenVazquez, last month • Primera entrega: proyecto limpio ...
45 app.UseMiddleware<CorrelationIdMiddleware>();
46 app.UseSerilogRequestLogging(); .....
47 app.UseMiddleware<ExceptionMiddleware>(); .....
48
49 app.UseSession();
50 app.UseAuthorization();
51
52 app.MapControllerRoute(
53     name: "default",
54     pattern: "{controller=Home}/{action=Index}/{id?}");
55
56 await app.RunAsync();
57
```

12. Código del Middleware de CorrelationId

C# CorrelationIdMiddleware.cs

Middleware > C# CorrelationIdMiddleware.cs > ...

```
You, last week | 1 author (You)
1 using System;
2 using System.Threading.Tasks;
3 using Microsoft.AspNetCore.Http;
4 using Serilog.Context;
5
6 You, last week | 1 author (You)
7 namespace VulnerableApp.Middleware
8 {
9     2 references | You, last week | 1 author (You)
10    ....public class CorrelationIdMiddleware
11    ....{
12
13        2 references
14        ....private readonly RequestDelegate _next;
15
16        0 references
17        ....public CorrelationIdMiddleware(RequestDelegate next) => _next = next;
18
19        0 references
20        ....public async Task InvokeAsync(HttpContext context)
21        ....{
22            ....var cid = Guid.NewGuid().ToString();
23            ....context.Response.Headers["X-Correlation-ID"] = cid;
24            ....using (LogContext.PushProperty("CorrelationId", cid))
25            ....{
26                ....await _next(context);
27            }
28        }
29    }
30 }
```

13. Código del Exception Middleware

C# ExceptionMiddleware.cs M

Middleware > C# ExceptionMiddleware.cs > ...

```
You, last week | 1 author (You)
1 using System;           You, last week • Simon
2 using System.Threading.Tasks;
3 using Microsoft.AspNetCore.Http;
4 using Microsoft.Extensions.Logging;
5
6 You, last week | 1 author (You)
7 namespace VulnerableApp.Middleware
8 {
9     4 references | You, last week | 1 author (You)
10    ....public class ExceptionMiddleware
11    ....{
12        2 references
13        ....private readonly RequestDelegate _next;
14        2 references
15        ....private readonly ILogger<ExceptionMiddleware> _logger;
16
17        0 references
18        ....public ExceptionMiddleware(RequestDelegate next, ILogger<ExceptionMiddleware> logger)
19        ....{
20        ....    _next = next;
21        ....    _logger = logger;
22
23        0 references
24        ....public async Task InvokeAsync(HttpContext context)
25        ....{
26        ....    try { await _next(context); }
27        ....    catch (Exception ex)
28        ....    {
29        ....        _logger.LogError(ex, "Excepción no controlada detectada");
30
31        ....        if (!context.Response.HasStarted)
32        ....        {
33        ....            context.Response.StatusCode = 500;
34        ....        }
35    }
36 }
```

14. Controlador instrumentado (ejemplo)

C# ApiController.cs

Controllers > C# ApiController.cs > ...

```

1  using Microsoft.AspNetCore.Mvc;
2  using VulnerableApp.Data;
3  using System.Diagnostics;
4
5  namespace VulnerableApp.Controllers
6  {
7      [ApiController]
8      [Route("api")]
9      public class ApiController : ControllerBase
10     {
11         private readonly ApplicationDbContext _db;
12         private readonly ILogger<ApiController> _logger;
13
14         public ApiController(ApplicationDbContext db, ILogger<ApiController> logger)
15         {
16             _db = db;
17             _logger = logger;
18         }
19
20         [HttpGet("user/{id}")]
21         public IActionResult GetUser(int id)
22         {
23             var watch = Stopwatch.StartNew();
24             var userAgent = HttpContext.Connection.RemoteIpAddress?.ToString();
25             var currentUser = User.Identity?.Name ?? "Anónimo";
26
27             _logger.LogInformation("Entrada a GetUser. ID solicitado: {Id}. Usuario: {User}, IP: {IP}", id, currentUser, userAgent);
28
29             try
30             {
31                 var currentUserId = HttpContext.Session.GetInt32("UserId");
32
33                 if (currentUserId == null || currentUserId != id)
34                 {
35                     _logger.LogWarning("Intento de acceso no autorizado al ID: {Id} por usuario: {User}", id, currentUser);
36                     return Unauthorized(new { message = "Acceso denegado." });
37                 }
38
39                 var user = _db.Users.Find(id);
40                 if (user == null)
41                 {
42                     _logger.LogWarning("Usuario con ID {Id} no encontrado", id);
43                     return NotFound();
44                 }
45
46                 watch.Stop();
47                 // 4. Registro de Salida con Tiempo
48                 _logger.LogInformation("Salida de GetUser. Tiempo: {Tiempo}ms", watch.ElapsedMilliseconds);
49
50                 return Ok(new { user.Id, user.Username, user.Email });
51             }
52             catch (Exception ex)
53             {
54                 // 5. Registro de Error
55                 _logger.LogError(ex, "Error crítico en GetUser para ID: {Id}", id);
56                 throw;
57             }
58         }
59     }
60
61

```

Conclusiones

La instrumentación de logging implementada en VulnerableApp permitió capturar un volumen representativo de 8,588 eventos que cubren de forma consistente el ciclo de vida de las solicitudes HTTP, los eventos de seguridad simulados (SQL Injection, XSS e IDOR) y las excepciones controladas de la aplicación.

El análisis en Seq confirmó que los mecanismos de detección funcionan correctamente: los 20 intentos de SQL Injection y los 30 intentos de XSS quedaron perfectamente identificables mediante los mensajes de log generados en SearchController y CommentController, y los 230 intentos de acceso no autorizado a recursos de la API quedaron documentados como advertencias.